



INSPIRE FORENSICS

(Recognised by the Indian Society of Forensic Science)

CUET-UG 2026

SAMPLE PAPER

A National Level Socio-Forensic Voluntary Student Led Educational Firm

CHEMISTRY (306)

Q1. The van't Hoff factor for NaCl in aqueous solution is nearly: (PYQ CUET UG 2022)

- A. 1
- B. 2
- C. 3
- D. 4

Q2. Which of the following is a colligative property? (PYQ NEET 2021)

- A. Viscosity
- B. Osmotic pressure
- C. Surface tension
- D. Refractive index

Q3. The molarity of pure water at 25°C is approximately:

- A. 1 M
- B. 18 M
- C. 55.5 M
- D. 100 M

Q4. Elevation in boiling point depends upon:

- A. Nature of solute
- B. Number of solute particles
- C. Nature of solvent only
- D. Chemical nature of solute

Q5. Which solution will have the highest osmotic pressure?

- A. 0.1 M glucose**
- B. 0.1 M NaCl**
- C. 0.1 M CaCl_2**
- D. 0.1 M urea**

**Q6. Oxidation takes place at which electrode in an electrochemical cell?
(PYQ CUET UG 2022)**

- A. Cathode**
- B. Anode**
- C. Salt bridge**
- D. Electrolyte**

Q7. SI unit of electrical conductivity is:

- A. ohm**
- B. siemens m^{-1}**
- C. volt**
- D. ampere**



Q8. Which cell converts chemical energy into electrical energy?

- A. Electrolytic cell**
- B. Galvanic cell**
- C. Fuel cell**
- D. Concentration cell**

**Q9. The standard electrode potential of hydrogen electrode is taken as:
(PYQ NEET 2020)**

- A. +1.00 V**
- B. -1.00 V**
- C. 0.00 V**
- D. +0.50 V**

Q10. In electrolysis of molten NaCl, sodium is liberated at:

- A. Anode**
- B. Cathode**
- C. Both electrodes**
- D. Salt bridge**

Q11. Unit of rate constant for a first-order reaction is: (PYQ CUET UG 2022)

- A. $\text{mol L}^{-1} \text{s}^{-1}$**
- B. s^{-1}**
- C. $\text{L mol}^{-1} \text{s}^{-1}$**
- D. $\text{mol}^2 \text{L}^{-2} \text{s}^{-1}$**

Q12. Half-life of a first-order reaction depends on: (PYQ NEET 2020)

- A. Initial concentration**
- B. Rate constant**
- C. Temperature only**
- D. Pressure**



Q13. The order of reaction is determined experimentally from: (PYQ NEET 2020)

- A. Balanced equation**
- B. Rate law**
- C. Stoichiometry**
- D. Molecularity**

Q14. For a zero-order reaction, rate is independent of:

- A. Time**
- B. Temperature**
- C. Concentration**
- D. Catalyst**

Q15. Increase in temperature increases reaction rate because:

- A. Activation energy increases**
- B. Collision frequency increases**
- C. Enthalpy increases**
- D. Equilibrium shifts**

Q16. Which element shows maximum number of oxidation states? (PYQ NEET 2019)

- A. Fe**
- B. Mn**
- C. Cu**
- D. Zn**

Q17. Colour of transition metals is due to:

- A. d–d transition**
- B. s–p transition**
- C. Charge transfer**
- D. Nuclear transition**



Q18. Lanthanides show contraction due to:

- A. Poor shielding of 4f electrons**
- B. High nuclear charge**
- C. Increase in atomic size**
- D. Presence of d electrons**

Q19. Which ion is colourless?

- A. Cu^{2+}**
- B. Mn^{2+}**
- C. Zn^{2+}**
- D. Fe^{3+}**

**Q20. Which transition metal is used as catalyst in Haber process?
(PYQ NEET 2021)**

- A. Copper**
- B. Iron**
- C. Nickel**
- D. Platinum**

Q21. Most stable oxidation state of lanthanides is:

- A. +2**
- B. +3**
- C. +4**
- D. +5**

Q22. Coordination number of central metal ion in $[\text{Fe}(\text{CN})_6]^{3-}$ is:

- A. 4**
- B. 6**
- C. 2**
- D. 3**



Q23. Ligand which donates more than one pair of electrons is called:

- A. Monodentate**
- B. Bidentate**
- C. Ambidentate**
- D. Polydentate**

Q24. Which of the following is a chelating ligand? (PYQ NEET 2020)

- A. NH_3**
- B. H_2O**
- C. EDTA**
- D. Cl^-**

Q25. Werner's theory explains:

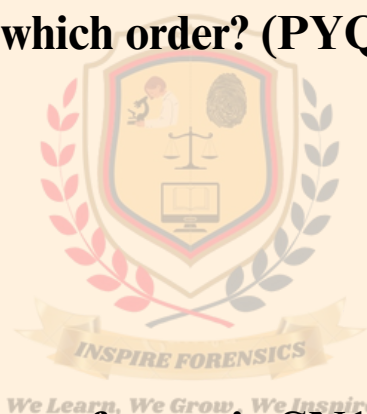
- A. Bonding in organic compounds**
- B. Isomerism in coordination compounds**
- C. Metallic bonding**
- D. Ionic bonding**

Q26. Which reaction is used to prepare alkyl halide from alcohol?

- A. Dehydration**
- B. Substitution**
- C. Elimination**
- D. Oxidation**

Q27. SN1 reaction follows which order? (PYQ NEET 2019)

- A. Zero order**
- B. First order**
- C. Second order**
- D. Third order**



Q28. Which alkyl halide reacts fastest in SN1 reaction?

- A. CH_3Cl**
- B. $\text{C}_2\text{H}_5\text{Cl}$**
- C. $(\text{CH}_3)_3\text{CCl}$**
- D. $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$**

Q29. Aryl halides are less reactive than alkyl halides due to: (PYQ CUET UG 2023)

- A. Partial double bond character**
- B. Steric hindrance**
- C. High electronegativity**
- D. Resonance only**

Q30. Which reagent is used in Wurtz reaction? (PYQ NEET 2020)

- A. Zn**
- B. Na**
- C. Mg**
- D. Al**

Q31. Chlorobenzene does not undergo S_N1 reaction because:

- A. Strong C–Cl bond**
- B. Resonance**
- C. sp^2 hybridisation**
- D. All of the above**

Q32. Functional group of alcohol is: (PYQ CUET UG 2023)

- A. $-\text{COOH}$**
- B. $-\text{OH}$**
- C. $-\text{CHO}$**
- D. $-\text{CO}-$**



Q33. Phenol is acidic due to:

- A. Inductive effect**
- B. Resonance stabilization**
- C. Hyperconjugation**
- D. Steric effect**

Q34. Which compound gives iodoform test? (PYQ NEET 2020)

- A. Methanol**
- B. Ethanol**
- C. Phenol**
- D. Ether**

Q35. Dehydration of alcohol leads to formation of: (PYQ CUET UG 2022)

- A. Alkane**
- B. Alkene**
- C. Aldehyde**
- D. Acid**

Q36. Williamson synthesis is used to prepare:

- A. Alcohol**
- B. Ether**
- C. Aldehyde**
- D. Ketone**

Q37. Phenol reacts with NaOH to form:

- A. Sodium phenoxide**
- B. Sodium ethoxide**
- C. Sodium benzoate**
- D. Benzene**

b. Maximum in situation (B)

c. Maximum in situation (C)

d. The same in all situations



Q38. Order of acidity is: (PYQ NEET 2019)

- A. Alcohol > Phenol**
- B. Phenol > Alcohol**
- C. Ether > Phenol**
- D. Ether > Alcohol**

Q39. Ether has which type of linkage?

- A. C–C**
- B. C–O–C**
- C. C–N**
- D. O–O**

Q40. Functional group of aldehyde is:

- A. -COOH**
- B. -CO-**
- C. -CHO**
- D. -OH**

Q41. Tollens' reagent is used to test: (PYQ NEET 2019)

- A. Ketones**
- B. Aldehydes**
- C. Acids**
- D. Alcohols**

Q42. Which compound is most acidic?

- A. Alcohol**
- B. Phenol**
- C. Carboxylic acid**
- D. Aldehyde**



Q43. Oxidation of aldehyde gives:

- A. Alcohol**
- B. Ketone**
- C. Carboxylic acid**
- D. Alkene**

Q44. Amines are basic in nature due to: (PYQ: CUET UG 2023)

- A. Lone pair on nitrogen**
- B. Presence of H atom**
- C. Resonance**
- D. Inductive effect only**

Q45. Which amine is most basic in aqueous solution? (PYQ NEET 2020)

- A. NH_3**
- B. Primary amine**
- C. Secondary amine**
- D. Tertiary amine**

Q46. Glucose is an example of: (PYQ CUET UG 2023)

- A. Protein**
- B. Lipid**
- C. Carbohydrate**
- D. Vitamin**

Q47. Which bond links amino acids in proteins? (PYQ NEET 2019)

- A. Glycosidic**
- B. Peptide**
- C. Ester**
- D. Hydrogen**



Q48. DNA contains which sugar? (PYQ CUET UG 2022)

- A. Ribose**
- B. Glucose**
- C. Deoxyribose**
- D. Fructose**

Q49. Vitamins required in small quantities are called:

- A. Macronutrients**
- B. Micronutrients**
- C. Enzymes**
- D. Hormones**

Q50. Which is a reducing sugar? (PYQ NEET 2021)

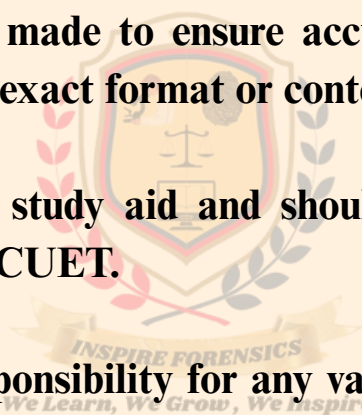
- A. Glucose**
- B. Sucrose**
- C. Starch**
- D. Cellulose**



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Answer Key

Question number	Answer Key		Question number	Answer Key
1	B		19	A
2	C		20	B
3	A		21	C
4	D		22	D
5	B		23	A
6	C		24	B
7	A		25	C
8	D		26	A
9	B		27	D
10	C		28	B
11	A		29	C
12	D		30	A
13	B		31	B
14	C		32	C
15	A		33	D
16	B		34	A
17	D		35	B
18	C		36	C

Answer Key

Question number	Answer Key
37	A
38	D
39	B
40	C
41	A
42	D
43	B
44	C
45	A
46	D
47	B
48	C
49	A
50	D